3rd SPGC

Department of Sustainable Energy Engineering
Indian Institute of Technology Kanpur

Proposal for Course Modification

Course Title		Introduction to Sustainable Energy Policy
Number		SEE-617
Credits (L-T-P [C])		3-0-0-9
Departments proposing the course	:	Sustainable Energy Engineering
Name of the Proposer	:	Deepika Swami, Rajeev Jindal
Offered for		PG Students of SEE and other departments 3/4th year UG students of other departments
Status of the course	:	PG Elective/ UG Elective
Prerequisite(s) for the course	:	Consent of Instructor
Faculty members interested in teaching	:	Deepika Swami, Rajeev Jindal, Sheo Shankar Rai, Indu Shekhar Chaturvedi, Anna Agarwal
Other Departments/Programmes of whose the students are expected to take up the course	:	ME, EE, CHE, MSE, AE, IME, ECO

Objective

The course will introduce different aspects of energy policy including energy resources, energy security, energy efficiency, energy justice, energy transition, integration of energy policy with climate governance, challenges and opportunities associated with coal and renewable energy. One of the major requirements is that energy policy should be aligned towards our development and rapid economic growth. Policymakers have predominately relied on policy instruments that utilize price mechanisms to deploy sustainable energy without giving much consideration to security and transition concerns and larger geopolitical questions of *who* owns the energy system, and *how* the energy needs to be generated and distributed.

Expected Learning Outcomes

The students completing this course will be equipped with the theoretical and analytical abilities to examine the nuanced relationship between energy policy and climate change and how the transition towards sustainable energy systems will change the overall scenario of energy system. They will understand the role of various policy instruments and institutions in shaping India's energy policy. The course will develop analytical and critical thinking and introduce new perspectives about energy, environmental and climate policies at global and national level and will give a new dimension to their innovative thinking, motivating them to come up with policy relevant technical solutions in the field of sustainable energy.

Course outline

The course begins with providing an overall context about the history of energy system and introduction of India's energy landscape. Next, it focuses on understanding energy security, geopolitical factors important to achieve energy security for India. Following this, we will understand the production, generation and distribution of various energy resources and the instruments that can help in achieving smooth transition from fossil-based sources to renewable energy. Lastly, we will understand how global climate governance and commitments will shape India's energy future.

Lecture -wise break up

Sr. No.	Topics	Lectures
1	**Overview of the Indian Energy Scenario:** • Historical Development of India's and Global Energy Sector • Introduction to India's Energy Landscape: Energy Production, Demand, Supply, and Consumption Patterns • Linking energy and environment: Emissions, Emission factor, Role of energy, Kaya identity and need for energy transition	6
2	**Energy security, and formulation of energy policy** • India's energy landscape: growth and challenges • Energy security: challenges and role of policies • Formulation of energy policy: Major stakeholders and process involved	6
3	**Conventional energy sources and Indian energy portfolio** • Role of Fossil fuel sources: Role of coal, oil and gas in India's energy mix, including production, energy generation, environmental implication. Policy scenario and how is it evolving? • Indian economy and dependence on coal: Political economy of coal • India's energy management system – roles of GENCO, TRANSCO, DISCOM • Electricity act and its development	10

4	**India's energy transition and various policy initiatives** • Drivers of energy transition: Environmental, Social, Economic, and Technological Factors • Growth of non-fossil based: Wind, Solar, Bioenergy, Tidal, Geothermal. Current status and policy support. • Driving demand and generation growth through policy framework: RPO, GEOA, roof top solar, PM-KUSUM Yojna, NBP, PMUY, CAFC, EBP, EV, • Regulatory institutions (MNRE, SERC, BEE, CEA etc.) • Economics of energy generation and distribution: Feed-in Tariffs, PPAs, Tax Incentives, Subsidies, Grants, Carbon Pricing, Cap-and-Trade Systems, and Emission Trading Schemes • Integration of renewables in grid: policy framework • India's energy policy for critical minerals for renewable energy	10
5	**Indian energy policy and global climate governance** • National and international missions: UNFCCC, COPs, NPACC, NCAP • India's commitments and agreements (Kyoto, Paris, Glasgow) • India's efforts in integrating renewable energy policies with climate change mitigation strategies • Future of energy generation in context of climate change	8

References

Books

1. Gupta, J. (2014). Handbook of Climate Change and India: Development, Politics and Governance.

2. Laird, F. N. (2004). Power politics: equity and environment in electricity reform: Washington, DC: World Resources Institute, 2002. xv+ 175 pp. Paperback. $30.00. *Utilities Policy, 12*(1), 51-51.

Papers

Chapman, A.J., McLellan, B.C., Tezuka, T., 2018. Prioritizing mitigation efforts considering co-benefits, equity and energy justice: Fossil fuel to renewable energy transition pathways. Appl. Energy 219, 187–198. https://doi.org/10.1016/j.apenergy.2018.03.054

Dubash, N.K., Ghosh, S., 2019. National Climate Policies and Institutions. India a Warm. World 329–348. https://doi.org/10.1093/oso/9780199498734.003.0019

Dubash, N.K., Joseph, N.B., 2016. Evolution of Institutions for Climate Policy in India. Econ. Polit. Wkly. 51, 44–54.

Fischer, F., Miller, G.J., 2017. Handbook of public policy analysis: theory, politics, and methods. Routledge.

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International Energy Agency, 2020. India 2020: Energy Policy Review, International Energy Agency. https://doi.org/10.1007/BF03404634

Keady, W., Panikkar, B., Nelson, I.L., Zia, A., 2021. Energy justice gaps in renewable energy transition policy initiatives in Vermont. Energy Policy 159, 112608. https://doi.org/10.1016/j.enpol.2021.112608

Kumar, G.S., 2017. Anatomy of Indian energy policy: A critical review. Energy Sources, Part B Econ. Plan. Policy 12, 976–985. https://doi.org/10.1080/15567249.2017.1336814

NITI Aayog, 2017. Draft National energy policy, NITI Aayog.

Patlitzianas, K.D., Doukas, H., Kagiannas, A.G., Psarras, J., 2008. Sustainable energy policy indicators: Review and recommendations. Renew. Energy 33, 966–973. https://doi.org/10.1016/j.renene.2007.05.003

Shukla, P.R., Chaturvedi, V., 2012. Low carbon and clean energy scenarios for India: Analysis of targets approach. Energy Econ. 34, S487–S495. https://doi.org/10.1016/j.eneco.2012.05.002

Tongia, R., Gross, S., 2019. Coal in India: Adjusting to Transition. Brookings Pap. 7.

Course proposed by	**Recommended/ Not recommended**	**This course is approved/not approved**
(Deepika swami)	Convener, DPGC (SEE)	Chairman, SPGC
(Rajeev Jindal)		

Modidifed course vs. present course: Revised course looks at the energy policy in a more holistic manner, with consideration of various dimensions including, energy resources, their exploration, production, generation along with the broad range of policy initiatives related to both renewable and non-renewable energy, which were found to be missing in the earlier version of the course. Thereby, this course is more comprehensive in nature and focuses on integrating technological, environmental, economic and social aspects with energy policy. The revision in course content is > 50%,

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Proposal for a New Course This is the current version

Course Title		Introduction to Sustainable Energy Policy
Number		SEE-617
Credits (L-T-P [C])		2-0-0-6
Departments proposing the course	:	Sustainable Energy Engineering
Name of the Proposer	:	Pradip Swarnakar Anoop Singh
Offered for		PG Students of SEE and other departments 3/4th year UG students of other departments
Status of the course	:	PG Elective/ UG Elective
Prerequisite(s) for the course	:	Consent of Instructor
Faculty members interested in teaching	:	Pradip Swarnakar, Anoop Singh
Other Departments/Programmes of whose the students are expected to take up the course	:	ME, EE, CHE, MSE, AE, IME, ECO

Course Objectives (Please outline objectives of the course)
The overall objective of the course is to provide students with a comprehensive understanding of the energy policy in India with special reference to clean and renewable energy sourcesthe role of sustainable energies in shaping the future trajectories of the policy. With climate change threatening the very survival of humanity, the energy transition has climbed to the top of the priority list for the policy-makers. The transition is a multi-dimensional process with technical and social transformations at the heart of it. The course will discuss role of policy framework in shaping up this transition.

Expected Learning Outcomes (Please outline what will students be able to achieve after doing the course)
The students completing this course will be equipped with the theoretical and analytical abilities to examine the nuanced relationship between energy policy and society and how the transition towards sustainable energy systems affects and modifies this relationship. They will understand how the technological innovations are embedded in the overall socio-political fabric. The course will give a human and social dimension to their innovative thinking, motivating them to come up with socially sensitive and policy relevant technical solutions in the field of sustainable energy.

Course outline

The course begins with providing an overall context about the relationship between energy and society. Next, it focuses on theoretical approaches to understand the policy processes. Following this, it traces the historical development of the energy policy in India. The major emphasis of the course will be on the more contemporary development of the energy policy in India within the context of global climate change governance and the increasing prominence of sustainable energies within it. The course ends with foregrounding the policy challenges of governing a just sustainability transition towards renewables.

Lecture wise breakup

S.No.	Topics	Lectures
1.	Introduction: Situating energy within social relations and society	02
2.	Public Policy: Definition; various approaches to study public policy – elite theory, systems theory, group theory, institutional theory, rational-choice theory	03
3.	Energy Policy in India (1): (a) Historical background and overview;(b) Understanding the dependence on of conventional sources: the post-independence importance of coal in the Indian context; the socio-political economy of coal and its integration into the energy policy.	03
4.	Energy Policy in India (2): Institutional development and major regulatory bodies. Sector-specific policy discussion, coal, oil, electricity, RE etc.	02
5.	Indian energy policy and global climate governance (pre-2007): Claims on Energy justice and insulation from global climate politics; India's role in building the global Southern block	01
6.	Indian energy policy and global climate governance (post 2007): Increased global integration of energy policy with global climate governance; role of renewables in national plan against climate change: NAPCC, Copenhagen 2009 and Paris Agreement 2015. Understanding energy policy instruments within the context of climate change: coal tax and subsidies of renewables.	04
7.	Renewable energy and challenges for Energy Policy in India: Governing energy transition efficiently- how to price renewables; how to make them financially viable for mass acceptance; The role of science and technology in assisting policy making around energy.	03
8.	Renewable energies and contested futures: Understanding just transitions: Why we need justice in renewable energy transition?	02
9.	Environmental Policy Action in India: Coal tax/cess, Energy efficiency and energy security, Energy access and renewable energy, Sustainable energy resources (challenges and opportunities)	06

Text-books, reference books, suggested readings and any other references

Dubash, Navroz K. 2019. India in a Warming World: Integrating Climate Change and Development. Oxford University Press.

Dubash, Navroz K., Radhika Khosla, Narasimha D. Rao, and Ankit Bhardwaj. 2018. "India's Energy and Emissions Future: An Interpretive Analysis of Model Scenarios." Environmental Research Letters 13(7):074018. doi: 10.1088/1748-9326/aacc74.

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Singh, A. (2009). A market for renewable energy credits in the Indian power sector. Renewable and Sustainable Energy Reviews, 13(3), 643-652.

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Swarnakar, Pradip. 2019. "Climate Change, Civil Society, and Social Movement in India." Pp. 253–72 in India in a Warming World. Oxford University Press.

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Course proposed by	**Recommended/ Not recommended**	**This course is approved/~~not~~ approved**
(Pradip Swarnakar)	Convener, DPGC (SEE)	Chairman, SPGC

(Anoop Singh)